FRONTEND | CONCEPTS & DEMO

Short speaking guide based on the implemented local application

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Project: NoteSphere - Smart Student Notes Sharing Platform

Your 45-60 second opening

"We are responsible for the frontend of NoteSphere. We use Next.js with React and TypeScript to build the student and admin portals. Tailwind CSS and a custom global CSS file provide the layout, spacing, colors and responsive styles. React state handles forms, search filters, loading indicators and feedback. The UI calls our backend APIs instead of connecting directly to the database. PDF.js displays protected notes inside the browser. Our focus is a consistent interface that students can use easily on desktop and mobile."

What technology is used, and why?

TECHNOLOGY / FILE	HOW TO EXPLAIN IT
Next.js 16 + React 19	Next.js provides routes and page rendering. React builds reusable UI components such as note cards, forms, sidebar and portal screens.
TypeScript / TSX	TypeScript adds compile-time type checks. TSX lets us describe HTML-like UI inside React components. The browser ultimately receives HTML, CSS and JavaScript.
Tailwind CSS 4 + globals.css	Tailwind is imported into src/app/globals.css. That file also holds custom styles, CSS variables, media queries and theme rules. The project does not use a file named style.css.
Lucide + PDF.js	Lucide supplies icons. PDF.js renders PDF pages with navigation, zoom and fullscreen controls.
Client state + API calls	React hooks manage local state and fetch data. Async requests update screens without treating browser state as database storage.

Divide the live explanation

• **Harmandeep:** Explain the landing page, layout, reusable components, sidebar, responsive CSS and light/dark styling.

• **Aahaan:** Explain student/admin forms, search/filter interactions, validation messages, loading/empty states and the PDF viewer.

Show: switch portals, resize a page, search as a student, open a PDF and save a bookmark. Show the resulting toast or library update.

FRONTEND | QUESTIONS & CODE MAP

Use these answers after rehearsing the actual code and demonstration

Why use React and Next.js together?

React is the component library. Next.js is the framework around it that provides routing, server rendering and backend route handlers in the same application.

Is Node.js part of the frontend?

Node.js runs development/build tools and the server. Frontend components run in the browser when they need interaction; Node.js is not the CSS framework or the database.

How is the page responsive?

CSS flex/grid layouts, media queries and reusable spacing rules adapt the interface to smaller screens. The sidebar can collapse, and controls wrap or stack as space reduces.

Does hiding the Upload button secure the application?

No. Hiding it improves the student experience. The backend separately checks the authenticated role and rejects student upload requests, even if someone changes the URL or request.

Where are the important frontend files?

All paths below are relative to outputs/notesphere/ in the delivered project.

src/components/app.tsx
Application shell, navigation and landing page.

src/components/auth.tsx
Student/admin sign-in and registration screens.

src/components/pages.tsx + src/components/ui.tsx
Functional pages, shared UI, data fetching and toast feedback.

src/components/pdf-viewer.tsx + src/app/globals.css
PDF reading controls and the central stylesheet.

Closing line: "Our UI is connected to real API responses; bookmarks, reviews and collections are saved in the database, not just shown as temporary cards."